

Processed tart cherry products--comparative phytochemical content, in vitro antioxidant capacity and in vitro anti-inflammatory activity.

Processing of fruits and vegetables affects their phytochemical and nutrient content. Tart cherries are commercially promoted to possess antioxidant and anti-inflammatory activity. However, processing affects their phytochemical content and may affect their related health benefits. The current study compares the in vitro antioxidant capacity and anti-inflammatory cyclooxygenase activity of processed tart cherry (*Prunus cerasus*) products—cherry juice concentrate, individually quick-frozen cherries, canned cherries, and dried cherries. Cherry products were analyzed for total anthocyanin and proanthocyanidin content and profile. On a per serving basis, total anthocyanins were highest in frozen cherries and total proanthocyanidins were highest in juice concentrate. Total phenolics were highest in juice concentrate. Juice concentrate had the highest oxygen radical absorbance capacity (ORAC) and peroxynitrite radical averting capacity (NORAC). Dried cherries had the highest hydroxyl radical averting capacity (HORAC) and superoxide radical averting capacity (SORAC). Processed tart cherry products compared very favorably to the U.S. Dept. of Agriculture-reported ORAC of other fresh and processed fruits. Inhibition of in vitro inflammatory COX-1 activity was greatest in juice concentrate. In summary, all processed tart cherry products possessed antioxidant and anti-inflammatory activity, but processing differentially affected phytochemical content and in vitro bioactivity. On a per serving basis, juice concentrate was superior to other tart cherry products. © 2012 Institute of Food Technologists®